

Object Oriented Programming with Java

Mid Term Examination Solutions (E1PY201B)

Section A: Java Architecture & Fundamentals

1. Explain what bytecode is and why it is important in Java programming.

Bytecode is the intermediate representation of Java source code that has been compiled by the Java compiler (javac). It is stored in `.class` files. **Importance:** It is highly optimized and platform-independent. This allows Java to follow the "Write Once, Run Anywhere" (WORA) principle, as the same bytecode can execute on any device equipped with a Java Virtual Machine (JVM).

2. Explain the execution of a Java program.

The execution involves three main stages:

1. **Compilation:** Source code (`.java`) is converted into bytecode (`.class`) by the compiler.
2. **Loading:** The Class Loader brings the bytecode into memory.
3. **Execution:** The JVM interprets or uses JIT (Just-In-Time) compilation to convert bytecode into machine-specific instructions for the underlying hardware.

Section B: Arrays & Exception Handling

4. Analyze the Enhanced For Loop code snippet.

The provided code uses a nested enhanced for loop to traverse a 2D array `{{1, 2}, {3, 4}}`.

Output: 1 2 3 4

Justification: The outer loop iterates through each row (1D array), and the inner loop iterates through each integer value within that row, printing them sequentially.

5. Error vs. Exception and Multiple Catch Blocks.

- **Error:** Indicates serious problems that a reasonable application should not try to catch (e.g., `OutOfMemoryError`, `StackOverflowError`).
- **Exception:** Conditions that a reasonable application might want to catch (e.g., `IOException`, `NullPointerException`).

Multiple Catch Blocks: Java allows handling different exceptions in different ways using multiple catch blocks. **Rule:** More specific exceptions (subclasses) must be caught before more general exceptions (superclasses) to avoid "unreachable code" errors.

Section C: Advanced Concepts & Programming

6. Explain Java Packages.

A package is a mechanism used to group related classes, interfaces, and sub-packages.

- **Types:** Built-in (e.g., `java.util`) and User-defined.
- **Creation:** Use the `package` keyword at the very top of the source file.
- **Access:** Using the `import` keyword or the fully qualified class name.

7. Merge Sorted Arrays Program.

```
public static int[] mergeSorted(int[] A, int[] B) {  
    int[] C = new int[A.length + B.length];  
    int i = 0, j = 0, k = 0;  
    while (i < A.length && j < B.length) {  
        if (A[i] <= B[j]) C[k++] = A[i++];  
        else C[k++] = B[j++];  
    }  
    while (i < A.length) C[k++] = A[i++];  
    while (j < B.length) C[k++] = B[j++];  
    return C;  
}
```